

Table of Contents

Preface	x
What I found, and why it took a book	x
How this book is built	xi
What this book is not	xii
A word on the series, and on honesty	xii
Who I hope reads it	xiii
Conventions Used in This Book	xiii
Using Code Examples	xiv
Thanks	xiv
 Part I. The Machine That Decides	 1
1. One Small Step	2
1.1 The Oldest Trick in Computing	4
1.2 Programs as State Transitions	5
1.3 Everything Is Secretly a Loop	7
1.4 What “Frozen” Means	8
1.5 Where Guarantees Have Traditionally Come From	9
1.6 The First Crack in the Ice	10
1.7 Lab: A Loop You Can Trust	11
Where We’ve Landed	15
What’s Next	15
 2. The Ghost in the Loop	 17
2.1 One Line of Difference	18
2.2 A Policy, Not an Oracle	19
2.3 Two Species of Wrong	20
2.4 How to Make a Poet Fill In a Form	21
2.5 The Shell Stays Frozen	22
2.6 G(body), Named	23
2.7 Lab: Melting the Meridian Loop	24
Where We’ve Landed	29
What’s Next	30

3. What the Machine Knows.....	31
3.1 Five Kinds of State.....	33
3.2 What Counts as an Observation.....	34
3.3 Who Told You That?.....	35
3.4 The Common Envelope.....	36
3.5 The World Moves On.....	37
3.6 Lab: Three Sources, One Truth.....	38
Where We've Landed.....	41
What's Next.....	41
4. The Art of the Next Move.....	42
4.1 The Decision Surface.....	43
4.2 Planning versus Acting, and the Price of Thinking.....	45
4.3 Acting: Where the Loop Touches the World.....	46
4.4 The Loop Closes: Feedback as New State.....	47
4.5 Lab: The Meridian ODA Runtime.....	49
Where We've Landed.....	51
What's Next.....	52
Part II. Loops in the Wild.....	53
5. A Field Guide to Loops.....	54
5.1 The Common Ancestor.....	55
5.2 The Tool-Use Loop.....	56
5.3 The Coding Loop.....	57
5.4 The Research Loop.....	58
5.5 The Verification Loop.....	59
5.6 Choosing Your Species.....	60
5.7 Lab: The Meridian Research Loop.....	61
Where We've Landed.....	64
What's Next.....	64
Part III. The Craft.....	66
6. Prompts Are Policies.....	67
6.1 The Anatomy of a Policy.....	69
6.2 Modularity and Versioning.....	70
6.3 Regression-Testing a Paragraph.....	71
6.4 Injection: A Policy Failure Before a Security Failure.....	72
6.5 Portability.....	73
6.6 Lab: A Paragraph on Trial.....	74
Where We've Landed.....	76
What's Next.....	77

7. The Instruction Set	78
7.1 The Analogy, Taken Seriously	80
7.2 Granularity: The RISC Question	81
7.3 The Contract	82
7.4 Privilege and Sandboxes	83
7.5 Discovery in a Fixed Catalog	84
7.6 Watching the Instruction Stream	85
7.7 Lab: Two Menus	85
Where We've Landed	88
What's Next	89
8. Working Memory	90
8.1 The Anatomy of the Window	92
8.2 Assembly: The Render Grows Up	93
8.3 Retrieval	94
8.4 Summarization and Compression	94
8.5 Poisoning and Isolation	96
8.6 Budgets and Observability	97
8.7 Lab: Two Renders	97
Where We've Landed	100
What's Next	100
9. The Exoskeleton	101
9.1 The Model Is Not the Agent	102
9.2 The Suit, Assembled	103
9.3 Harness Patterns	104
9.4 Portability	105
9.5 Lab: The Signature Build	106
Where We've Landed	108
What's Next	109
Part IV. Keeping It Honest	110
10. Knowing When to Stop	111
10.1 The Two Deaths	112
10.2 The Stopping Stack	113
10.3 The Proposal Rule	115
10.4 Stopping as a Design Object	116
10.5 Lab: The Word for Enough	116
Where We've Landed	119
What's Next	119

11. When Loops Go Wrong	121
11.1 The Shape of the Bestiary	123
11.2 Failures of Action	123
11.3 Failures of Motion	123
11.4 Failures of the Picture	125
11.5 Failures of the Objective	125
11.6 Loops Meeting Loops	126
11.7 The Recovery Ladder	127
11.8 Lab: The Pathology Bench	129
Where We've Landed	131
What's Next	131
12. Trust, but Verify	133
12.1 The Verifier Ladder	134
12.2 Judges and Their Calibration	135
12.3 The Gate as Teacher	137
12.4 Evaluation: From Outputs to Trajectories	137
12.5 The Threat Model	138
12.6 Defenses That Do Not Ask Nicely	139
12.7 The Verification Debt Ledger	141
12.8 Lab: The Incident, Twice	141
Where We've Landed	145
What's Next	145
13. The Price of a Thought	147
13.1 What a Step Costs	148
13.2 Meter One: The Price of Depth	149
13.3 Meter Two: The Value of One More Try	150
13.4 Meter Three: The Honest Denominator	151
13.5 The Fourth Currency	152
13.6 Lab: Reading the Meter	153
Where We've Landed	155
What's Next	156
Coda. The Agent Is Still Frozen	157
The margin, collected	158
What an agent actually is	159
The zoo	159
The question	160
A. The Minimal Agent Runtime	162
Running it	162
What is where	163

The fourteen lines that swap the model.....	164
Extending it.....	164
What it deliberately does not do.....	165
The File, Complete.....	165
B. Tool Schema Patterns.....	175
The six-field entry.....	175
Naming.....	175
Argument schemas.....	176
Descriptions, which are read at runtime.....	176
Error contracts.....	176
Contract tests.....	176
C. State Schema Patterns.....	177
The common envelope.....	177
The five kinds of state.....	177
Freshness and reconciliation.....	178
Serialization.....	178
D. Context Engineering Checklist.....	179
E. Evaluation Recipes.....	181
Recipe 1: the charter battery.....	181
Recipe 2: growing the battery from incidents.....	181
Recipe 3: trajectory properties.....	181
Recipe 4: judge calibration.....	182
Recipe 5: fleet telemetry.....	182
Recipe 6: cost evaluation.....	182
F. Security Checklist.....	183
G. Observability Event Schema.....	185
H. Production Readiness Checklist.....	186
I. The Labs, Complete.....	188
Lab 1: A Loop You Can Trust.....	188
Lab 2: Melting the Meridian Loop.....	191
Lab 3: Three Sources, One Truth.....	195
Lab 4: The Meridian ODA Runtime.....	198
Lab 5: The Meridian Research Loop.....	204
Lab 6: A Paragraph on Trial.....	210
Lab 7: Two Menus.....	214
Lab 8: Two Renders.....	220
Lab 9: The Signature Build.....	223

Lab 10: The Word for Enough.....	229
Lab 11: The Pathology Bench.....	233
Lab 12: The Incident, Twice.....	236
Lab 13: Reading the Meter.....	243